Ans. (d) : Abid Hussain Committee - 1997 on Small Scale Industries
Mahajan Committee - Sugar Industry 1997
Bhagwati Committee - Unemployment
Chelliah Committee - Tax Reforms
Narasimham Committee - Financial Reforms

67. The new series of Wholesale Price Index (WPI) released by the Government of India is with reference to the base prices of:
(a) 1981-82 (b) 1990-91
(c) 1993-94 (d) 1994-95

Ans. (c) : When the question was asked Base year for WPI was 1993-94. Earlier 1981-82 was being used as the base year currently it is 2011-12.

68. The term National Income represents:
(a) Gross national product at market prices minus depreciation
(b) Gross national product at market prices minus depreciation plus net factor income from abroad
(c) Gross national product at market prices minus depreciation and indirect taxes plus subsidies
(d) Gross national product at market prices minus net factor income from abroad

Ans. (c) : Net National Product at Factor Cost (NNPFC) is called the national income of a country.
Net national product = (GDP + net income earned from foreign countries).

69. Assertion (A) : There was an increase in industrial production during 1999-2000.
Reason (R) : The period witnessed a stable exchange rate and improved business sentiments.
(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true but R is NOT the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Ans. (a) : The development of industries has started in India only after 1980. The rapid economic growth that began after the 1991 economic liberalization is still on the path. The industrial growth rate during 1999-02 was 6.7%.

70. Assertion (A) : Ceiling on foreign exchange for a host of current account transaction heads was lowered in the year 2000.
Reason (R) : There was a fall in foreign currency assets also.
(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true but R is NOT the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Ans. (c) : This market where foreign exchange practices take place, the foreign exchange market and the rate at which it happens is called the foreign exchange rate. It varies in different regions.
In 2000, the foreign exchange rate expired on current account transactions.
Presently, there is a steady increase in the foreign exchange reserves.

PHYSICS

71. When light waves pass from air to glass, the variables affected are:
(a) Wavelength, frequency and velocity
(b) Velocity and frequency
(c) Wavelength and frequency
(d) Wavelength and velocity

Ans. (d) : Upon refraction from one medium to another, only the wavelength and velocity change, while frequency remains the same.
Option D is the correct one.

72. When water is heated from 0° C to 10° C, its volume
(a) Increases
(b) Decreases
(c) Does not change
(d) First decreases and then increases

Ans. (d) : The highest density of water is at 4°C. Unusual spread is seen in water. On heating the water, its volume decreases to 0°C to 4 °C. Thereafter, it starts increasing.

73. Match List I with List II and select the correct answer using the code given below the lists:

List I (Characteristic)	List II (Particle)
A. Zero mass	1. Positron
B. Fractional charge	2. Neutrino
C. Fractional spin	3. Quark
D. Integral spin	4. Photon

	A	B	C	D
(a)	4	3	1	2
(b)	3	2	4	1
(c)	2	3	4	1
(d)	3	2	1	4

Ans. (a) : Quarks have fractional electric charge values–either 1⁄3 or 2⁄3 times the elementary charge. The positron has an electric charge of +1e, a spin of 1/2, and has the same mass as an electron. A neutrino is an electrically neutral, weakly interacting elementary subatomic particle with half integer spin.

74. The mass of a body on earth is 100 kg (acceleration due to gravity, g_o = 10 m/s^2). If acceleration due to gravity on the moon is (g_e/6), then the mass of the body on the moon is
(a) 100/6 kg (b) 60 kg
(c) 100 kg (d) 600 kg

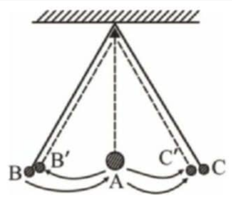
Ans. (c) : Mass is a universal constant. The mass of a body remains unchanged in any part of universe. Mass of a body does not change with respect to gravity. It is the weight that changes with gravity.

75. Who amongst the following was the first to state that the earth was spherical?
(a) Aristotle (b) Copernicus
(c) Ptolemy (d) Strabo

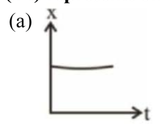
Ans. (a) : Aristotle in 340 B.C first stated that the earth was spherical in his book "On the Heaven".

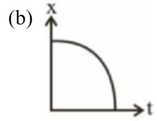
76. Consider the following statements:
A simple pendulum is set into oscillation. Then
1. The acceleration is zero when the bob passes through the mean position.
2. In each cycle the bob attains a given velocity twice.
3. Both acceleration and velocity of the bob are zero when it reaches its extreme position during its oscillation.
4. The amplitude of oscillation of the simple pendulum decreases with time.
Which of these statements are correct?
(a) 1 and 2 (b) 3 and 4
(c) 1, 2 and 4 (d) 2, 3 and 4

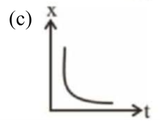
Ans. (c) :

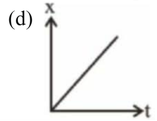


The bob of a simple pendulum is attached to a string which pulls the bob along its length. Here B and C are extreme positions whereas A is mean position. The speed of the bob increases as it approaches the mean position A and continues to move till it reaches C. At C the speed becomes zero. Due to the unbalanced force the bob moves towards the mean position. The speed of the bob is maximum at the mean position and is zero at the extreme positions. Thus, it is clear that in each cycle bob velocity increases from zero to maximum. This means that it attains a given velocity twice. Suppose the bob of the pendulum reaches up to B while oscillating, then AB is the amplitude. For the next oscillation the bob fails to reach B but it will reverse the direction from point BC' instead of B. The amplitude of oscillation in the second case is ABC' which is less than AB. That means, a retarding force is acting on the bob thereby reducing the amplitude of oscillation. This retarding force is nothing but air-resistance or air-friction. At extreme position, acceleration is maximum. So statement 3 is not correct.

77. Which one of the following distance-time graph (x-t) represents one-dimensional uniform motion?
(a) 

(b) 

(c) 

(d) 

Ans. (d) : In one dimensional uniform motion equal distance is travelled at equal interval of time. So velocity is constant and hence acceleration is zero. So graph (d) represents one dimensional uniform motion.

78. Assertion (A) : A stick is dipped in water in a slanting position. If observed sideways, the stick appears short and bent at the surface of water.
Reason (R) : The light coming from the stick undergoes scattering from water molecules giving the stick a short and bent appearance.
(a) Both A and R are individually true and R is the correct explanation of A
(b) Both A and R are individually true but R is NOT the correct explanation of A
(c) A is true but R is false
(d) A is false but R is true

Ans. (c) : When a ray of light passes from one medium to another, it deflects from its path. When it goes from sparse to dense medium, it goes towards the normal and away from the opposite position.

CHEMISTRY

79. Which one of the following is not radioactive?
(a) Astatine (b) Francium
(c) Tritium (d) Zirconium

Ans. (d) : Out of the given options zirconium is not radioactive. Generally elements which have atomic number above 80, shows radioactivity. However Tritium (atomic no. 1) is radioactive isotope of hydrogen due to presence of a high number of neutrons compared to the number of protons.

80. Consider the following.
In a nuclear reactor, self-sustained chain reaction is possible, because.
1. More neutrons are released in each of the fission reactions.
2. The neutrons immediately take part in the fission process.
3. The fast neutrons are slowed down by Graphite.
4. Every neutron released in the fission reaction initiates further fission.